THE WORK OF ELLENBERG AND VENKATESH ON BOUNDS FOR ℓ-TORSION OF CLASS GROUPS

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1. Introduction

Let $K = \mathbb{Q}(\sqrt{D})$, let Cl_K be the ideal class group of K, and let

$$\mathrm{Cl}_K[\ell] \coloneqq \{a \in \mathrm{Cl}_K : a^\ell = I\},$$

be the ℓ-torsion subgroup of Cl_K, where I is the identity in Cl_K. Since $\#\mathrm{Cl}_K \ll_\varepsilon D^{1/2+\varepsilon}$, we always have the trivial bound $\#\mathrm{Cl}_K[\ell] \ll_\varepsilon D^{1/2+\varepsilon}$.

Previously, we have seen the following:

Theorem 1 (Pierce [3])**.** *Let $K = \mathbb{Q}(\sqrt{D})$ for any square-free D. We have*

$$\#\mathrm{Cl}_K[3] \ll_\varepsilon D^{1/2-1/56+\varepsilon}.$$

Which we saw follows from the square sieve. We have also seen the following:

Theorem 2 (Helfgott and Venkatesh [2])**.** *Let $K = \mathbb{Q}(\sqrt{D})$ for any square-free D. We have*

$$\#\mathrm{Cl}_K[3] \ll D^{0.4417\ldots}.$$

This follows from height theory and sphere packing to bound the number of integral points on an elliptic curve.

Today, we will look at the work of Ellenberg and Venkatesh [1], where they prove the following:

Theorem 3 (Ellenberg and Venkatesh [1])**.** *Let $K = \mathbb{Q}(\sqrt{D})$ for any square-free D. We have*

$$\#\mathrm{Cl}_K[3] \ll D^{1/3+\varepsilon}.$$

More generally, for any $\ell \in \mathbb{N}$, we have

$$\#\mathrm{Cl}_K[\ell] \ll D^{1/2-1/2\ell+\varepsilon}.$$

2. Preliminaries

The main idea of the proof is fairly elementary, though there will be some heavy lifting done from the reflection principle, and generalisations also require some substantial adjustments.

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Some algebra. We recall some preliminary definitions and facts to get started.

Definition (Norms)**.** *Let K be a field and let L be a finite extension of K. If L/K is a Galois extension, then the* ***field norm*** *is defined to be*

$$N_{L/K}(\alpha) \coloneqq \prod_{\sigma \in \mathrm{Gal}(L/K)} \sigma(\alpha).$$

It can also be defined more generally as the determinant of the linear transformation that scales by α. If L is a number field with ring of integers $\mathcal{O}_L$, then the ***ideal norm*** *is defined to be*

$$N(I) \coloneqq |\mathcal{O}_K / I|.$$

Key facts: For L a number field, we have the following important facts:

1) For any square-free D, with $L = \mathbb{Q}(\sqrt{D})$ we have

$$N_{L/\mathbb{Q}}\big(x + y\sqrt{D}\big) = x^2 - Dy^2.$$

Importantly, the field norm can be negative if $D > 0$. Geometrically, fixed norm defines a circle for imaginary quadratic fields, and a hyperbola for real quadratic fields .

2) If $\mathfrak{a} = (a)$ is a principal ideal of $\mathcal{O}_L$, then

$$N(\mathfrak{a}) = |N_{L/\mathbb{Q}}(a)|.$$

3. Using small split primes to upper bound the ℓ-torsion

We will give the proof in full in the case where we consider $K = \mathbb{Q}(\sqrt{-D})$, where $D \equiv 1, 2 \bmod 4$ is positive and square-free. This was outlined in Joseph's introductory talk. The case where $D \equiv 3 \bmod 4$ follows almost identically.

Claim 1. *Suppose that $x + y\sqrt{-D} \in \mathcal{O}_K$ and $\mathfrak{J} = (x + y\sqrt{-D})$ is a principal integral ideal. If $N(\mathfrak{J}) < D$, then it must be an extension of an ideal for $\mathbb{Q}$.*

Proof. Since $N(\mathfrak{J}) = x^2 + Dy^2$, we have

$$N(\mathfrak{J}) < D \Rightarrow y = 0, \text{ hence } \mathfrak{J} = (x).$$

Therefore, if $\mathfrak{J}$ is principal **and** $N(\mathfrak{J}) < D$, then it must be an extension of an ideal for $\mathbb{Q}$. □

Remark. *To handle the case where $D \equiv 3 \bmod 4$, one should reduce the bound D to $D/4$, since then $y \in \frac{1}{2}\mathbb{Z}$.*

We will crucially use the fact that such ideals are invariant under Galois conjugation.

Claim 2. *Suppose that p, q are distinct primes in $\mathbb{Z}$ that split in $\mathbb{Q}(\sqrt{-D})$, and $e, f \in \mathbb{Z}$ are such that $p^{|e|}q^{|f|} < D$, then P^eQ^f does not belong to the trivial class in Cl_K.*

Proof. By our assumptions, we have $(p) = P\overline{P}$ and $(q) = Q\overline{Q}$. If e, f are such that $p^eq^f < D$, then $N(P^eQ^f) < D$ (using the fact that the norm is multiplicative), so by Claim 1, P^eQ^f is principal, and for some $x \in \mathbb{Q}$, we have

$$(x) = P^eQ^f.$$

But taking Galois conjugates, we quickly see that this cannot be the case, so $\mathfrak{J}$ *cannot* be principal. □

The upshot is that, if P^eQ^f is principal in the class group, then the exponents e, f must be large. In particular, we can use this to say that P, Q represent the different elements modulo ℓ-torsion so long as $pq \leq D^{\frac{1}{2\ell}}$.

Claim 3. *Suppose that p, q are distinct primes in $\mathbb{Z}$ that split in $\mathbb{Q}(\sqrt{-D})$ so that $(p) = P\overline{P}$ and $(q) = Q\overline{Q}$. Then suppose that $p, q \leq D^{\frac{1}{2\ell}}$. Then P, Q represent different elements of the class group modulo ℓ-torsion*

Proof. Suppose that P and Q represent the same elements modulo ℓ-torsion, so that $[P] = [Q][I]$ in $\mathrm{Cl}_K/\mathrm{Cl}_K[\ell]$. It follows that $[PQ^{-1}] \in \mathrm{Cl}_K[\ell]$, and hence P^lQ^{-l} belongs to the trivial class in Cl_K. In lieu of Claim 2, this cannot be the case, since $p^lq^l < D$ by assumption. □

This claim tells us that counts for small split primes gives a lower bound for the number of residue classes modulo ℓ-torsion, which in turn gives an upper bound on the size of $\mathrm{Cl}_K[\ell]$.

Lemma (Quantification)**.** *Suppose that $D > 0$, let $K = \mathbb{Q}(\sqrt{-D})$, and let $\delta < 1/2\ell$. Suppose that $\{p_1, \dots, p_M\}$ are split primes in K with $N_{K/\mathbb{Q}}(p_i) \leq D^\delta$. Then $\#\mathrm{Cl}_K[\ell] \ll_{\varepsilon,\ell} D^{1/2+\varepsilon}M^{-1}$.*

Proof of Lemma. By Claim 3, we know that $p_1, \dots, p_M$ represent different residue classes modulo ℓ torsion. Therefore, we have $|\mathrm{Cl}_K/\mathrm{Cl}_K[\ell]| \geq M$. Seeing as

$$\#\mathrm{Cl}_K[\ell] = \frac{|\mathrm{Cl}_K|}{|\mathrm{Cl}_K/\mathrm{Cl}_K[\ell]|},$$

and using the fact that $\#\mathrm{Cl}_K \ll D^{1/2+\varepsilon}$, we deduce that $\#\mathrm{Cl}_K[\ell] \ll_{\varepsilon,\ell} D^{1/2+\varepsilon}M^{-1}$. □

The fix: extending to arbitrary number fields K. As mentioned at the beginning, if $D > 0$, we have no lower bound on the norm of elements of $\mathbb{Q}(\sqrt{D})$. For example, in $\mathbb{Q}(\sqrt{2})$, we have $N(1+\sqrt{2}) = -1$. In fact, $\varepsilon = 1+\sqrt{2}$ is a fundamental unit, and the set of generators for a principal ideal (a), $\{\pm\varepsilon^k a\}_{k\in\mathbb{Z}}$, is infinite, discrete, and not compact. Since the norm is $\sigma_1(\varepsilon)\sigma_2(\varepsilon) = -1$, if one of these embeddings is large, then the other must be tiny.

The fix is to look at the **Arakelov class group**. This the usual class group by a "circle" (or torus) related to the regulator, making the space compact, and there is an associated height function that allows us to measure the size. This allows the authors to treat real and imaginary fields uniformly (it is a compactification of the normal class group, analogous to compactification of elliptic curves). The methods can also be generalised to higher degree extensions, allowing them to prove the following:

Proposition 1. *Suppose K/K_0 is an extension of number fields of degree d, let ℓ be a positive integer, and let $\delta < \frac{1}{2\ell(d-1)}$. Suppose that $\{\mathfrak{p}_1, \dots, \mathfrak{p}_M\}$ are prime ideals of norm at most $\mathrm{Norm}(\mathfrak{D}_{K/K_0})^\delta$ that are unramified and are not extensions of prime ideals from any proper subfield of K containing K_0. Then*

$$\#\mathrm{Cl}_K[\ell] \ll_{[K:\mathbb{Q}],\varepsilon,\ell} (\mathrm{disc}\, K)^{1/2+\varepsilon}M^{-1}.$$

What remains is to try and show that there exist many small split primes in $\mathbb{Q}(\sqrt{-D})$.

4. Counting small split primes

The Chebotarev density theorem. The Chebotarev density theorem serves as a (vast) generalisation of Dirichlet's Theorem for primes in arithmetic progressions.

Proposition 2. *Let K be an extension of $\mathbb{Q}$ of degree d. Assuming GRH, we have*

$$\#\mathrm{Cl}_K[\ell] \ll_{d,\varepsilon} (\operatorname{disc} K)^{1/2-\frac{1}{2\ell(d-1)}+\varepsilon}.$$

Sketch of proof. By an effective version of the Chebotarev density theorem of Lagarias and Odlyzko, for any $\eta > 0$, there exist $\gg (\operatorname{disc} K)^{\eta-\varepsilon}$ primes of $\mathbb{Q}$ of size $\leq (\operatorname{disc} K)^{\eta}$ with split completely in K. Taking $\eta = \frac{1}{2\ell} - \varepsilon$, we get $M = (\operatorname{disc} K)^{\frac{1}{2\ell}-2\varepsilon}$. □

An unconditional result.

Lemma 4.1 (The Scholz reflection principle)**.** *Let E be a number field and let $K = E(\sqrt{\eta})$ be a quadratic extension of E not containing $\sqrt{-3}$. Then*

$$\left|\mathrm{rank}\mathrm{Cl}_{E(\sqrt{-3\eta})}[3] - \mathrm{rank}\mathrm{Cl}_{E(\sqrt{\eta})}[3]\right| \ll_{[E:\mathbb{Q}]} 1$$

This is [1, Lemma 5].

Some ideas involved in the proof. We begin with a base field E, and we consider the extensions $K = E(\sqrt{\eta})$ and $\widehat{K} = E(\sqrt{-3\eta})$. We can then form $L = E(\sqrt{\eta}, \sqrt{-3})$, which is a quadratic extension of both K and $\widehat{K}$ since K does not contain $\sqrt{-3}$. We know that $\zeta_3 = e^{4\pi i/3} \in L$. By class field theory, the 3-rank of the class group corresponds to the number of unramified abelian cubic extensions of K, which, by Galois theory, correspond to the unramified cubic extensions of L coming from K. These can be counted using Kummer theory, since we know that $\zeta_3 \in L$. □

A consequence of this is that $\#\mathrm{Cl}_{E(\sqrt{-3\eta})}[3] \asymp \#\mathrm{Cl}_{E(\sqrt{\eta})}[3]$.

We will combine this result with Proposition 1 to prove the following:

Theorem. *For any D square-free, $K = \mathbb{Q}(\sqrt{D})$, we have*

$$\#\mathrm{Cl}_K[3] \ll D^{1/3+\varepsilon}.$$

By Proposition 1, bounding the size of the 3-torsion of $\mathbb{Q}(\sqrt{D})$ can be done by finding split primes $p < D^{1/6-\varepsilon}$.

Proof. First of all, suppose that $p \nmid 6D$ is inert in $\mathbb{Q}(\sqrt{-3})$. The divisibility condition implies that $p \nmid 3$, so p is unramified in $\mathbb{Q}(\sqrt{-3})$. Essentially, p is split in $\mathbb{Q}(\sqrt{-E})$ $\iff$ $p = x^2 + Ey^2$ $\iff$ E is a square modulo p. Therefore, to be inert, we must have that $\left(\frac{-3}{p}\right) = -1$ (this is generalised in the Kummer–Dedekind Theorem). By complete multiplicativity of the Legendre symbol, and since p is unramified in these fields, we must have either (i) p splits in $\mathbb{Q}(\sqrt{D})$, or (ii) p splits in $\mathbb{Q}(\sqrt{-3D})$. If p were inert (or split) in *both simultaneously*, this would imply that $\left(\frac{-3D}{p}\right) = \left(\frac{D}{p}\right)$, and subsequently that $\left(\frac{-3}{p}\right) = 1$.

It follows by Quadratic reciprocity and the first supplement that $\left(\frac{-3}{p}\right) = -1 \iff p \equiv 2 \bmod 3$. Therefore, by Dirichlet's Theorem, for some $d \in \{D, -3D\}$, the number of primes $p \leq |d|^{1/6-\varepsilon}$ such that p splits in $\mathbb{Q}(\sqrt{d})$ is $\gg |d|^{1/6-\varepsilon}$. Applying Proposition 1 with $\ell = 3$, $\delta = 1/6 - \varepsilon/2$. We deduce that $\#\mathrm{Cl}_{\mathbb{Q}(\sqrt{d})}[3] \ll |d|^{1/2-1/6+\varepsilon} \ll D^{1/3+\varepsilon}$.

By Scholz's reflection principle, we know that $\#\mathrm{Cl}_{\mathbb{Q}(\sqrt{D})}[3] \asymp \#\mathrm{Cl}_{\mathbb{Q}(\sqrt{-3D})}$. Since we know that one of these is bounded above by $D^{1/3+\varepsilon}$, the bound holds for $\#\mathrm{Cl}_K[3]$, so we obtain the result. □

References

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